Atomic Force Microscopy

A Comprehensive Review on Instrumentation & Implementation

Samriddha Ganguly

Department of Physics
Indian Institute of Science Education and Research Bhopal

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Outline

1. Introduction
2. Quantum Walks: What and Why?
3. Quantum Walks as Computational Tools
4. Digression: Adiabetic Quantum Computing
5. Applications
6. Conclusion
7. Further Reading on my Part

Representation of Physical Phenomena

- The Representation of Physical Objects and their Abstraction
- What does it mean to be a good theory?
- Modeling and Instantiation
- The Scientific Process

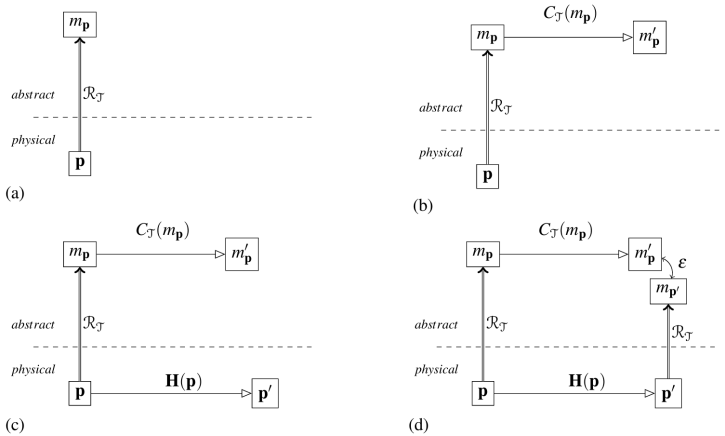


Figure: Abstraction and the Scientific Process (Kendon, 2020).

Computing

- We solve an abstract problem using a Physical System
- Key thing to note is that is instantiation is arbitrary. Different lead to the similar output if encoding and decoding is consistent.
- e.g.: binary and unary methods to encode and decode. This can be made precise using Group Theory
- A Computation requires *i*) Output *ii*) Representational Entity where the abstraction is physically instantiated.

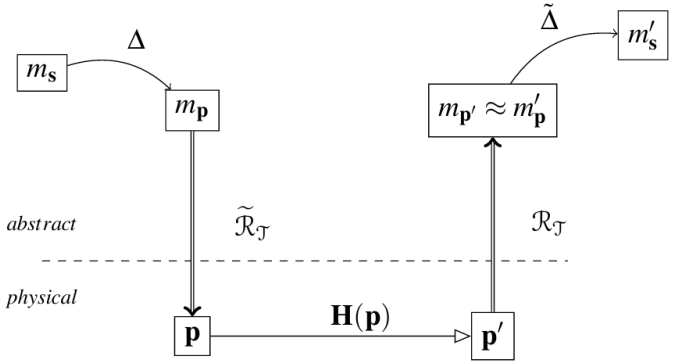


Figure: Physical Computer Diagram (Kendon, 2020).

Efficient Computing

1. Efficiency has different meanings for different purposes
   - Theorists define polynomial scaling of computational resources as being efficient
   - Practically it implies the fact that we obtain output in timescales relevant to us.
2. In Quantum Computing we look to exploit this practical definition and look to solve problems faster as compared to current classical computing.
3. Binary Encoding is one of the reasons why classical digital computers work efficiently. This is shown later as well that it works for Quantum Computing with qubits.
4. Physical Quantum Walks are difficult to work with like quantum walk on a lattice for purpose of computation of problems. However digital quantum walks through qubits is not far from realization!

What are Quantum Walks?

- Quantum analogue of classical random walks.
- Used for modeling transport, quantum algorithms, and more.
- Two types:
  - **Discrete-time** (coined walks)
  - **Continuous-time** (Hamiltonian-based)

Key Paper Focus

Continuous-time quantum walks for computation and their encoding into qubits.

Brief Overview of DTQW

1. Position Vector $\mathcal{H}_T = \{|j\rangle, j = 0, ..., n-1\}$ and Coin Vector $\mathcal{H}_C = \{|i\rangle, i = \uparrow, \downarrow\}$. The Hilbert Space for our Quantum Walk is thus $\mathcal{H}_T \otimes \mathcal{H}_C$
2. We may use "Coin Operators" like the Hadamard Coin

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

3. Shift Operator:

$$U = |\uparrow\rangle\langle\uparrow| \otimes \sum_j |j+1\rangle\langle j| + |\downarrow\rangle\langle\downarrow| \otimes \sum_j |j-1\rangle\langle j|$$

4. Total Evolution Operator incorporating randomness due to the coin operator is given by: $S = U(H \otimes I)$

CTQW: Core Formalism

1. Consider simplest case of a ring shaped graph with n vertices corresponding to each number state $|j\rangle$ in a Hilbert Space given by: $\mathcal{H}_T = \{|j\rangle, j = 0, ..., n-1\}$.
2. We find Laplacian of the Graph as $L = A - D$, where A is the adjacency Matrix and D is the diagonal matrix.
3. If L Symmetric define Hamiltonian H as $\langle j|H|k\rangle = -\gamma L_{jk}$
4. Thus evolution from an initial state $|\psi(0)\rangle$ can be given as follows: $|\psi(t)\rangle = \exp\{-iHt\}|\psi(0)\rangle$

Quantum Advantage

1. It was proven by Childs et al. (2009) that Quantum Walks are universal for quantum computing, restricted to single walkers encoded into qubits.
2. Later Childs (2013) proved that Multiple interacting quantum walkers provide universal quantum computing.
3. Bosonic Sampling provides midway- not requiring encoding in qubits but also not universal. Able to solve certain problems faster than best known classical algorithms.

Quantum Advantage Contd.

1. Searching in Unsorted database is one such case of Quantum Walks giving a quadratic speedup over classical counterparts. N vs $\sqrt{N}$.
2. Glued Tree algorithm (Childs et. al 2003) and Quantum Walk search are both permutation symmetric problems.
3. Useful for Testbed Algorithms to test Quantum Hardware

CTQW as an Algorithm

1. CTQW described before is not an algorithm.
2. We can use the glued trees algorithm approach to convert this to an algorithm.
3. Second Method is combining Hamiltonian for QW with another Hamiltonian that describes the problem where a marked state is treated differently from the other states. Both of these have to be encoded for efficient quantum computing.

Glued Tree Algorithm

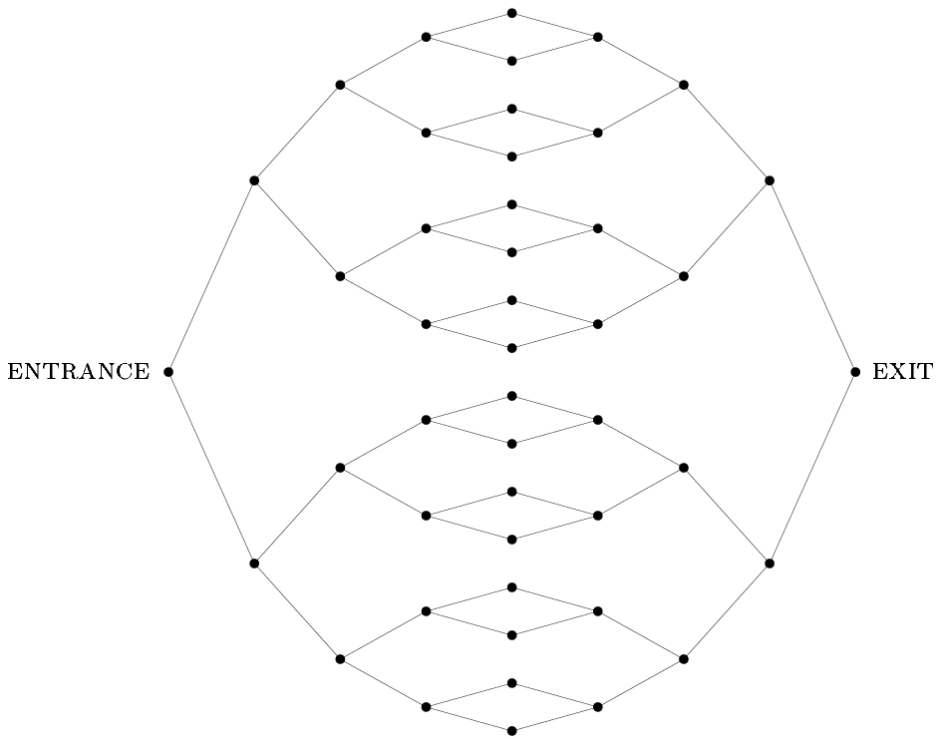


Figure: Glued Tree Algorithm: the problem is to find the exit node from the entrance node (the roots of the trees) offers an exponential speed up. (Childs et al. 2003)

Continuous Time Quantum Search

1. Here the problem is to find a marked state $|m\rangle$. So the problem Hamiltonian is $\hat{H}_m = \mathbf{1} - |m\rangle\langle m|$. $|m\rangle$ lower in energy.
2. In terms of Pauli Operators:

$$\hat{H}_m = \mathbf{1} - \frac{1}{2^n}\prod_{j=1}^{n}(\mathbf{1} + q_j\sigma_j^{(z)})$$

   . Here $q_j = -1, 1$ defines the bitstring. $-1 = 0$ to convert to bits.
3. $\sigma_j^{(z)}$ is the Pauli z-operator acting on jth qubit: $\sigma_j^{(z)}|1\rangle = -|1\rangle$ & $\sigma_j^{(z)}|0\rangle = |0\rangle$.

Encoding for Efficiency

Binary Encoding:

1. Hypercube: n qubits for 2^n vertices.
2. Exponential efficiency gain.
3. Now we will focus on the Quantum Walk Hamiltonian for this Hypercube. $\hat{H}_h = \gamma(n\mathbf{1} - \sum_{j=1}^{n} \sigma_j^{(x)})$
4. The $\sigma_j^{(x)}$ operator flips the jth qubit, similar to walking along the edges in the jth direction on the hypercube

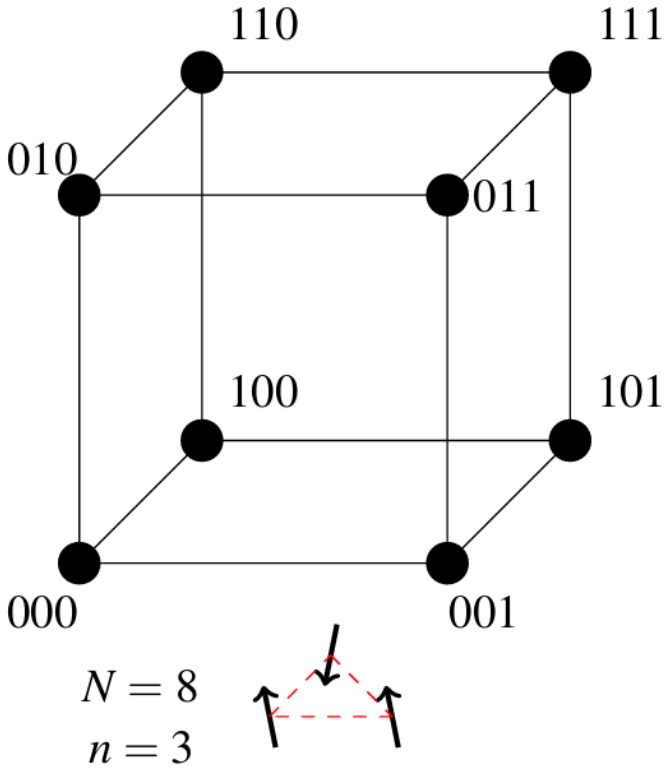


Figure: Hypercube graph and qubit encoding (Kendon, 2020).

Adiabetic Quantum Computing

1. The adiabetic theorem states that if time evolution is slow enough, the quantum system stays in the instantaneous eigenstate throughout the time evolution with high probability.
2. This "slow" depends on the energy gap
3. However considerations can help avoid pathological cases.
4. Idea is to evolve from a easy to prepare ground state to the problem Hamiltonian $\hat{H}_p$ smoothly from $\hat{H}_s$.
5. $\hat{H}_{AQC} = \{1 - s(t)\}\hat{H}_s + s(t)\hat{H}_p$, with $s(0) = 0$ & $s(t_f) = 1$

The Complete Quantum Search Algorithm

1. Now we can use the QW Hamiltonian for the Hypercube $\hat{H}_h$ and the marked state problem Hamiltonian $\hat{H}_m$, to find the combined Hamiltonian for the QWS Algorithm.

$$\hat{H}_{QWS} = \hat{H}_h + \hat{H}_m$$

2. The initial state is taken to be unbiased respecting our lack of knowledge:

$$|\psi(t=0)\rangle = 2^{-n/2} \sum_{j=0}^{2^n-1} |j\rangle = 2^{-n/2}(|0\rangle + |1\rangle)^{\otimes n}$$

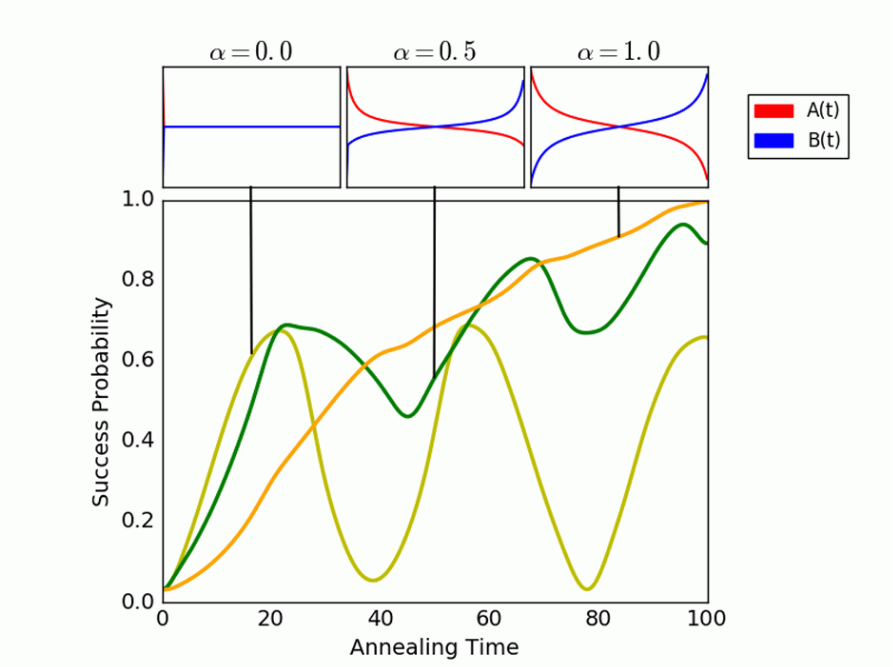
3. Acting $\hat{H}_{QWS}$ on it for a time $t_f \approx \pi\sqrt{N}/2$ results in a final state $|\psi(t_f)\rangle \approx |m\rangle$.
4. Problem is seen to be solved in $O(\sqrt{N})$, a quadratic improvement over classical best strategy.

Hybrid QW-AQC Algorithm

1. Both Quantum Walk Search and Adiabatic Quantum Computing are solvable in large N limit.
2. A Smooth interpolation between adiabatic and quantum walk can be done and optimal contributions from both mechanisms employed to improve performance on imperfect hardware.
3. important for understanding and implementation of quantum advantage in early Quantum Computers.

Hybrid QW-AQC Algorithm Contd.

hybrid continuous-time quantum search algorithms



interpolate between QW ($\alpha = 0$) and AQC ($\alpha = 1$)

$$\hat{H}(\alpha, t) = A(\alpha, t)\hat{H}_h + B(\alpha, t)\hat{H}_m$$

$$\hat{H}_{QW} = \gamma \hat{H}_h + \hat{H}_m$$

$$\hat{H}_{AQC} = [1 - s(t)]\hat{H}_h + s(t)\hat{H}_m$$

$\rightarrow$ need γ and $s(t)$ . . .

[James Morley's work (UCL CDT)
PRA 99, 022339 (2019)
arχiv:1709.00371]

Figure: Illustration of Hybrid QW-AQC Algorithms (Kendon 2020).

Hybrid Quantum Classical Algorithms: A natural path towards the future

1. Structured database and/or correlations are usually better without random guessing.
2. Studies by Montanaro (2018,2019) exhibit a hybrid discrete time quantum-classical algorithm which obtains a quantum advantage.
3. Can be done by locating bottleneck subroutines and exploiting them and replacing with quantum speed-up subroutines if possible.

Summary

- Quantum walks are powerful for both modeling and computation.
- Promise in trying Quantum Walks in problems where previously only Adiabatic Computing was used.
- Efficient encoding (e.g., hypercube) is critical for scalability.
- Coupling strengths between pairs of qubits, noise on quantum systems, no fully developed theory of error correction in continuous time setting- limit the Quantum Computing picture.
- Hybrid approaches (QW + AQC) show promise for near-term hardware.

Future Directions

Error correction, larger-scale implementations, and hybrid classical-quantum algorithms.

Topics to Ponder on

The following sections I was not able to grasp fully. I would like to read up on the background theory of the same.

1. Decision Trees, Graphs and Algorithms
2. Permutation Symmetric Problems
3. Adiabatic Quantum Computing from the Physical Point of view
4. Quantum Cellular Automata
5. Bosonic Sampling
6. Single Avoided Crossing Model
7. Sherrington-Kirkpatric spin glasses

References

- Viv Kendon (2020). *How to Compute Using Quantum Walks*. Durham University.
- Childs et al. (2009). *Universal computation by quantum walk*. Phys. Rev. Lett.
- Childs et al. (2003). *Exponential algorithmic speedup by a quantum walk*. Proc. 35th Annual ACM Symposium on Theory of Computing.
- E Farhi & S Gutmann (1998). *Quantum computation and decison trees* Phys. Rev. A.
- Ashley Montanaro (2018). *Quantum Walk speedup of backtracking algorithms*. Theory of Computing.

Thank You!